

Performance Testing

Date	30 JUNE 2026
Team ID	XXXXXX
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman, Swagger UI
Type of Testing	Performance Testing, Load Testing
Target Module / API	FastAPI Backend (/chat, /analyze-event, /fact-check)
Test Environment	Local FastAPI Server + Streamlit Frontend
Test Date	30 June 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Generate networking suggestions	10	60	Suggestions generated successfully
2	Analyze networking event details	20	60	Event analysis completed without errors
3	Verify facts using Wikipedia API	30	120	Fact verification returned correct results
4	Multiple users accessing chatbot simultaneously	50	180	System remains stable with acceptable response time

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Average Response Time	< 2 sec	1.5 sec	Pass	Fast response
2	Maximum Response Time	< 5 sec	3.8 sec	Pass	Within acceptable range
3	Throughput	10 Req/sec	12 Req/sec	Pass	Handled expected load
4	Error Rate	< 1%	0.3%	Pass	Very few errors
5	CPU Utilization	< 80%	62%	Pass	Stable CPU usage
6	Memory Utilization	< 80%	56%	Pass	Memory usage within limits

Step 4: Observations & Analysis

- The FastAPI backend handled multiple user requests successfully.
- Response time remained below the target during testing.
- Event analysis and fact verification worked correctly.
- No major crashes or failures occurred during testing.
- CPU and memory usage remained within acceptable limits.
- Overall, the Personalized Networking Assistant performed reliably under normal and moderate workloads.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

